Cairo University
Faculty of Science
Physics Departement

Superconductivity & The Josephson Effect

Student:
Abd El-Rahman Mostafa
Code: 2027063

Supervisor:
Dr. Abdelhamid Galal

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1 Abstract

Superconductivity and Josephson junctions are interesting phenomenons, the quasi-particles excitations in the first and the nonlinear chaotic behavior of the second, both are intriguing phenomenons leading to a lot of studying and research possibilities, the research for there applications shows a great promise for them to be essential in our future every day lives, As the potential to use Josephson arrays in Quantum computing and the idea of Room temperature superconductors can make a technological leap similar to that made by the steam engine. We model a system of three shunted series-connected Josephson junctions using RCSJ Model, We Present a numerical solution for the behavior of Three junctions connected in series shunted by a resistor, and try to do so with the tilted-washboard potential in mind.

2 Introduction to Superconductivity

2.1 Superconductivity Phenomena

The phenomena of superconductivity was discovered by Onnes in 1911, in simple terms it's the vanishing of material resistance below critical temperature specified for the material, in addition to critical temperature there is a critical magnetic field as high magnetic fields destroy superconductivity restoring normal conducting states. Superconductors have some special properties in addition to zero resistance (i.e,. Meissner effect[perfect diamagnetism], critical current, Josephson effect, flux quantization), also depending on the behavior of the superconductor during the phase transition due to critical magnetic field, superconductors are classified to type I and type II superconductors. Multiple theories and explanations have been proposed to explain the phenomena of superconductivity, giving a valuable prediction for physical quantities. Listing the theories according to timeline[4]:

1. London theory
2. Ginzbur-Landau Theory
3. BCS Theory

2.2 BCS: Bardeen-Cooper-Schrieffer Theory

In this Theory Superconductors are presumed to be a condensation of bound pairs of electrons. Using a thought experiment that if we introduce two new electrons to an already filled Fermi sea, the electrons might fall in to the Fermi sea if an attraction interaction could be formed between the two electrons, later this interaction turned out to be through electron-phonon interaction by the observation

of isotope effect in tin, where the critical temperature of tin isotopes (with different atomic mass) was inversely proportional to the square root of atomic mass.

In our effort to understand the behavior of the electron pairs, we utilize second quantization notation to write the ground state of the BCS theory, By the use of "Creation operators" $c^*_{k\sigma}$ which creates an electron of momentum k with spin σ, in addition the annihilation operators $c_{k\sigma}$ which empty the electron in the k state with σ, whereσ $=\uparrow,\downarrow$, these operators obey anti-commutation relations of fermion operators. Constructing the wave function of this system seems trivial by using a Grand-Canonical ensemble system and using a superposition of all possible electron pair states, this can be written by using the creation operators on the Fermi sea wave function.

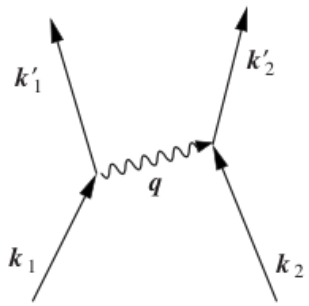

Figure 1: Feynman diagram of electron-electron interaction via phonon exchange[2].

$$|\psi_0\rangle = \sum_{k>k_F} g_k c^*_{k\uparrow} c^*_{-k\downarrow} |F\rangle \tag{1}$$

where $|F\rangle$ is the fermi-sea with all states filled up to k_F, the most general wave function for in the Grand-Canonical ensemble is by replacing the Fermi sea state with the vacuum state $|\phi_0\rangle$,

$$|\Psi_G\rangle = \prod_{k=k_l...k_m} (u_k + v_k c^*_{k\uparrow} c^*_{-k\downarrow})|\phi_0\rangle \tag{2}$$

So the ground state in BCS theory is a state of all cooper pairs constructed by superimposing wave functions of all pair states neglecting the interaction, where $|u_k^2| + |v_k^2| = 1$ making $|u_k^2|$ the propability the state is empty and $|v_k^2|$ is the probability that the state is occupied, it's more appropriate some times to use trigonometric identities.

$$u_k = sin\theta_k \tag{3}$$
$$v_k = cos\theta_k \tag{4}$$

Writing the Hamiltonian of the system is simply the summation of kinetic energy if each particle and the summation of every interaction that can annilate then create a pair of electrons, called the Pairing Hamiltoninan.

$$\mathcal{H} = \sum_{k\sigma} \epsilon_k n_{k\sigma} + \sum_{kl} V_{kl} c^*_{k\uparrow} c^*_{-k\downarrow} c_{-l\downarrow} c_{l\uparrow} \tag{5}$$

using the variational method we set $\xi_k = \varepsilon_k - E_F$ and calculate $\delta \langle \Psi_N | \mathcal{H} - N_{op}\mu | \Psi_N \rangle$ and solving Equation (6) to find the optimum value for

the wave function coefficients in terms of θ_k.

$$\frac{\partial}{\partial\theta_k}\langle\Psi_N|\mathcal{H}-N_{op}\mu|\Psi_N\rangle=0 \tag{6}$$

$$-2\xi_k\ Sin2\theta_k+\sum_l V_{lk}\ cos2\theta_k\ sin2\theta_k=0 \tag{7}$$

We define new quantities

$$\Delta_k=-\sum_l V_{kl}u_kv_l \tag{8}$$

$$E_k=(\Delta_k^2+\xi_k^2)^{1/2} \tag{9}$$

Where E_k is the excitation energy of a quasi-particle of momentum $\hbar k$ and Δ_k is the minimum excitation energy.
In Equation(5) the interaction term in the Hamiltonian is a Quadratic term, making the hamiltonian hard to solve, so using $\mathcal{M}$ean Field Theory approximation.

$$c_{-k\downarrow}c_{k\uparrow}=b_k+(c_{-k\downarrow}c_{k\uparrow}-b_k) \tag{10}$$

where $b_k=\langle c_{-k\downarrow}c_{k\uparrow}\rangle$, then Supstituting in the Hamiltonian we get.

$$\mathcal{H}=\sum_{k\sigma}\epsilon_k n_{k\sigma}+\sum_{kl}V_{kl}(c^*_{k\uparrow}c^*_{-k\downarrow}b_l+b^*_kc_{-l\downarrow}c_{l\uparrow}-b^*_kb_l) \tag{11}$$

we also define: $$\Delta_k=-\sum_l V_{kl}\langle c_{k\uparrow}c_{-k\downarrow}\rangle \tag{12}$$

Such a Hamiltonian can be diagonalized by linear transformation, defining new fermi operators γ_k

$$\gamma^*_{k0}=u^*_kc^*_{k\uparrow}-v^*_kc_{-k\downarrow} \tag{13}$$

$$\gamma^*_{k\uparrow}=u^*_kc^*_{-k\downarrow}+v^*_kc_{k\uparrow} \tag{14}$$

$$\gamma_{k0}=u_kc_{k\uparrow}-v_kc^*_{-k\downarrow} \tag{15}$$

$$\gamma_{k\uparrow}=u_kc_{-k\downarrow}+v_kc^*_{k\uparrow} \tag{16}$$

$$\tag{17}$$

γ^*_k operators create a quasi-particle excitations of two spin directions from the superconducting ground state, these particles are called Bogoliubons.
So much insights about the system could be presented by it's density of states, yet with the previous work one can predict the behavior of the system in finite temperature, calculate the critical temperature and other thermodynamic quantities as specific heat and internal energy, but to comprehend the phenomenon of tunneling in a simple way we choose to discuss it using the density of states.

2.3 Density of States

In finite temperature the probability of occupation of an excited state of energy E_k follows the Fermi function $f(E_k) = (e^{\beta E_k} + 1)^{-1}$, and as the excited fermion states have one-to-one correspondence relation to creation of electron states "c_k^*", the relation between normal and superconducting density of states is:

$$N_s(E)dE = N_n(\xi)d\xi \tag{18}$$

$$N_n(\xi) = n(0) \tag{19}$$

$$\frac{N_s(E)}{N(0)} = \frac{d\xi}{dE} = \begin{cases} \frac{E}{(E^2-\Delta_k^2)^{1/2}} & \text{if } E > \Delta_k \\ 0 & \text{if } E < \Delta_k \end{cases} \tag{20}$$

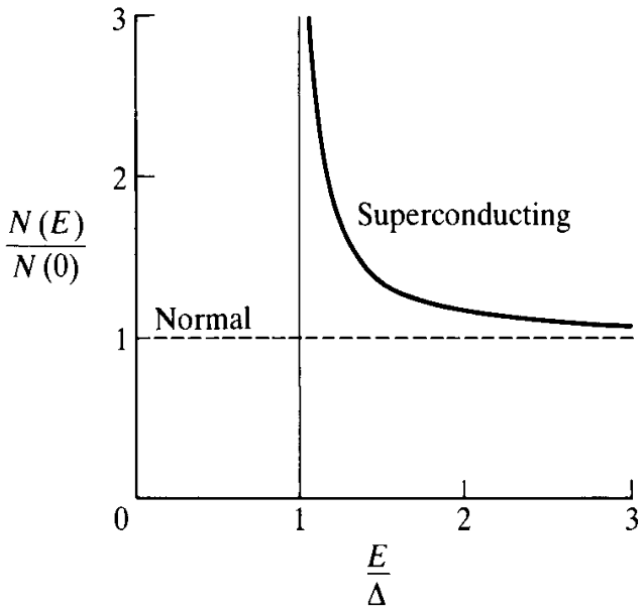

Figure 2: Density of states of superconducting compared to normal state, all **k** states with energy lying in the normal metal region are raised in energy above the gap in the superconducting state[1].

3 Electron Tunneling

The basic idea is that there is a non-zero probability that charge transfer by tunneling of electrons from a conductor to the other through a thin insulating barrier, the probability depends on the details of the insulating material. Using the energy level diagram for the SIS(superconductor insulator superconductor) junction one can understand the idea of tunneling, the diagram shows energies below and above the Fermi level. Explaining the

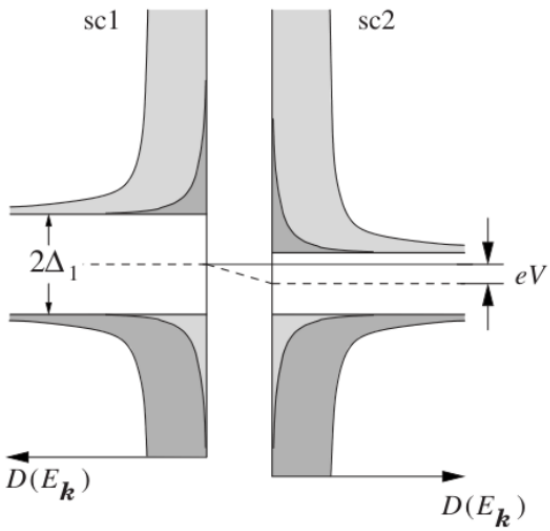

Figure 3: the diagram shows the energy levels when energy is applied Dark gray represents filled states and light gray represents empty states [2].

At zero temperature $T = 0$ the Fermi surfaces of the two superconductors are on the same level, so tunneling probability is zero, as there are no empty states that the electrons can occupy after tunneling. Applying a voltage over the junction supplies energy to the superconductor creating a hole on one side and a particle on the other, in simple words energy states elevate on one side and fall on the other yet there can be no tunneling if the electrons still don't have empty states on the other side of the material, until the applied voltage equals double the excitation energy of quasi-particles $eV = 2\Delta$.
For finite temperature $T \neq 0$ some electrons are excited to the upper energy levels leaving energy states empty for electrons to tunnel into.

4 Ginzburg-Landau Theory

Using landau Theory for second-phase transition, which expands the free energy in terms of an order parameter, that permits the theory to give good description at temperatures near critical temperature T_c, near interfaces or in the mixed state of Type II superconductor. Assuming the the wave function ψ is small and varies slowly in space, we use $|\psi|^2$ as the order parameter, we expand the free energy with the idea that $|\psi|^2 = n_s$.

$$f = f_{n0} + \alpha|\psi|^2 + \frac{\beta}{2}|\psi|_4 + \frac{1}{2m^*}|(\frac{\hbar}{i}\nabla - \frac{e^*}{c}A)\psi|^2 + \frac{h^2}{8\pi} \tag{21}$$

Where h is the magnetic field, for $\psi = 0$ the free energy returns to its normal state $f_n = f_{n0} + \frac{h^2}{8\pi}$, So in the absence of fields and gradients.

$$f_s - f_n = \alpha|\psi|^2 + \frac{\beta}{2}|\psi|_4 \tag{22}$$

finding the minimum of the free energy as a variable of $|\psi|^2$ we find that

$$|\psi|^2 = |\psi_\infty|^2 = -\frac{\alpha}{\beta} \tag{23}$$

where ψ_∞ conventionally means deep in the interior of the superconductor, our system implies that the order parameter $|\psi|^2$ tends to zero as we approach the transition temperature, where the free energy is lowest, this implies that β is positive such that minimum energy don't happen at very large $|\psi|^2$, leaving us with two cases where:

- α positive so minimum energy is at $\psi = 0$
- α negative so minimum satisfies Equation (23)

For negative α substituting the value of ψ_∞ in Equation (22) we find that.

$$f_s - f_n = \frac{-H_c^2}{8\pi} = -\frac{\alpha^2}{2\beta} \tag{24}$$

where H_c is the thermodynamic critical field.
The dependence of α on temperature is necessary, as it can help differentiate the behavior around the critical temperature,for $\alpha > 0$ we are describing the free energy in the normal state, and for bigger than zero we describe the free energy in superconducting state, as by definition critical temperature is the highest temperature at which $|\psi|^2 \neq 0$ this gives Ginzburg-Landau theory an edge describing superconductors at regions near critical temperature.

4.1 Ginzburg-Landau Differential Equations

For this paper application we discuss how the wave function behave when field currents or gradients apply over the superconductor, which appears to be the trivial idea that, $\psi(r) = |\psi(r)|e^{i\phi(r)}$ adjusts it's self to minimize the free energy of the system, So going back to Equation(21)

$$f = f_{n0} + \alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4 + \frac{1}{2m}|(\frac{\hbar}{i}\nabla - \frac{e^*}{c}A)\psi|^2 + \frac{h^2}{8\pi} \tag{25}$$

taking the first derivative and equating the equation to zero to see investigate the properties of the order parameter $|\psi|^2$.

$$\alpha|\psi| + \beta|\psi|^2\psi + \frac{1}{2m^*}|(\frac{\hbar}{i}\nabla - \frac{e^*}{c}A)^2\psi| = 0 \tag{26}$$

Introducing the normalized wave function $f = \frac{\psi}{\psi_\infty}$ where $\psi_\infty^2 = \frac{-\alpha}{\beta}$ and choosing the simple case of no fields ($A = 0$), so the equation becomes in one dimension.

$$\frac{\hbar^2}{2m^*|\alpha|}\frac{d^2f}{dx^2} + f - f^3 = 0 \tag{27}$$

and we define the characteristic length $\xi^2(T) = \frac{\hbar^2}{2m^*|\alpha(T)|}$, so the equation in terms of $\xi(T)$ is.

$$\xi^2(T)\frac{d^2f}{dx^2} + f - f^3 = 0 \tag{28}$$

This equation shows a very peculiar case, as any small disturbance for ψ from ψ_∞ will decay in a characteristic length of order $\xi(T)$, this can be seen by substituting $f(x) = 1 + g(x)$ in the Ginzburg-landau differential equation (28).

$$\xi^2 g" + 1 + g - 1 - 3g - \mathcal{O}(g^2) = 0 \tag{29}$$

Solving this equation neglecting higher order terms.

$$g" = (\frac{2}{\xi^2})g \tag{30}$$

$$g(x) \propto e^{\sqrt{2}x/\xi(T)} \tag{31}$$

Knowing this information we can advance to the Josephson junctions with complete understanding of the behavior of the macroscopic wave function at the boundaries.

5 Josephson Junction (JJ)

The DC Josephson effect is a phenomenon caused by tunneling of cooper pairs through an insulating barrier, this tunneling produces a super current without an applied voltage(zero voltage super-current).
A Josephson junction is a solid state physics device that exhibits the Josephson effect and consists of 2 superconducting electrodes separated by a thin insulating barrier,weak links between superconducting electrodes can be an insulating layer or a normal metal layer.
The link (or bridge) can be described by one dimensional Ginzburg-Landau differential Equation(28) as we mentioned before.

$$\xi^2(T)\frac{d^2f}{dx^2} + f - f^3 = 0 \tag{32}$$

We are assuming the electrodes are in equilibrium so $|f| = 1$ in both of them, but the phase of the order parameter may differ.
The behavior of the junction is given by the gauge invariant phase difference between the superconducting electrodes.

$$\varphi = \theta_1 - \theta_2 - \frac{2e}{\hbar}\int A.dl \tag{33}$$

Where the integration is from one electrode of the weak link to the other, and θ_i the phase of the macroscopic wave function in the electrode i is ($\Psi_i = \sqrt{n_i}e^{i\theta_i}$) where A is the vector potential.

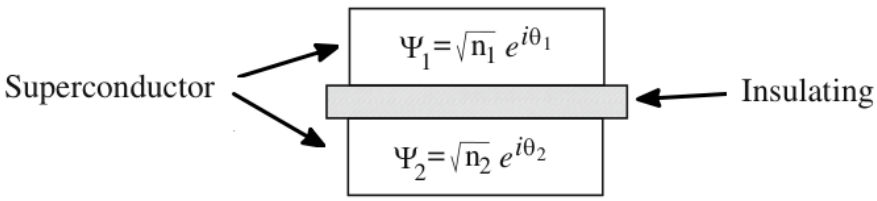

Figure 4: Schematic diagram of a Josephson junction showing the macroscopic wave function of the superconducting electrode[5].

The basic equations for the Josephson effect are:

$$I_s = I_c sin\ \phi \tag{34}$$

$$V = \frac{\hbar}{2e}\frac{d\phi}{dt} \tag{35}$$

These equations establish the DC Josephson effect, I_c is the critical current which is the maximum current the junction can reach with zero voltage across the junction(having constant phase), and I_s is the superconducting current.

5.1 The RCSJ Model

The critical current $I_c(T)$ is enough to characterize the zero voltage DC properties of a weak link, but for finite voltage involving AC Josephson effect a more complete description is required.
The RCSJ(resistively and Capacitive shunted junction) model speaks for it's self, as we model the ideal junction to shunted by resistance R and capacitance C.
Using the circuit diagram in Figure 5 the total current throught the junction is the sum of three components(Josephson super-current, resistance current and capcitive current).

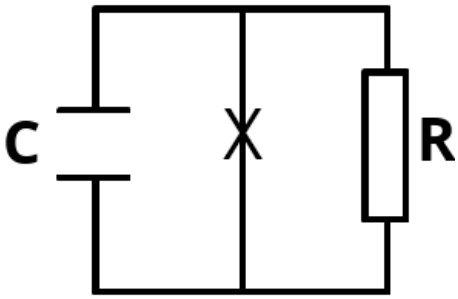

Figure 5: .

$$I = I_J + I_R + I_C \tag{36}$$

$$I = I_c \; sin\phi + \frac{1}{R}V + C\dot{V} \tag{37}$$

Normalizing the current with respect to critical current ($i = \frac{I}{I_c}$), and substituting using Equation (35)

$$i = sin \; \phi + Q^{-1}\dot{\phi} + \ddot{\phi} \tag{38}$$

The time variable in this equation is $\tau = \omega_p t$ which is dimensionless and $\omega_p = (\frac{2eI_c}{\hbar C})^{\frac{1}{2}}$ is called the plasma frequency of the junction, and the quality factor $Q = \omega_p RC$.

5.1.1 The Tilted-Washboard Model

A simplification for Equation(38) presented in [1] is that it's one-to-one mapping to a classical particle of mass $(\hbar/2e)^2 C$ moving on φ-axis in a potential.

$$U(\varphi) = -E_j cos \; \varphi - \frac{\hbar I}{2e}\varphi \tag{39}$$

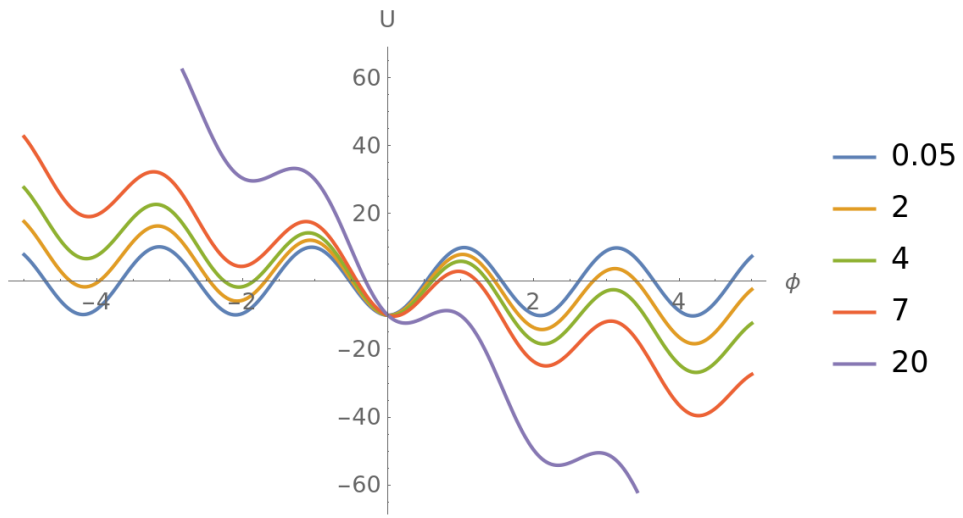

Figure 6: the graph shows the washboard potential for different values of I, It's clear that as the value of I increase the slope of the potential increase.

According to the quality factor there are two modes of operation for the junction:

- Over-damped junction, C is so small such that $Q \ll 1$ the phase moves more slowly corresponding to lower instantaneous voltage
- Under-damped junction, C is very large such that $Q \gg 1$, the I-V curve becomes hysteric as the behaviour change according to the initial condition.

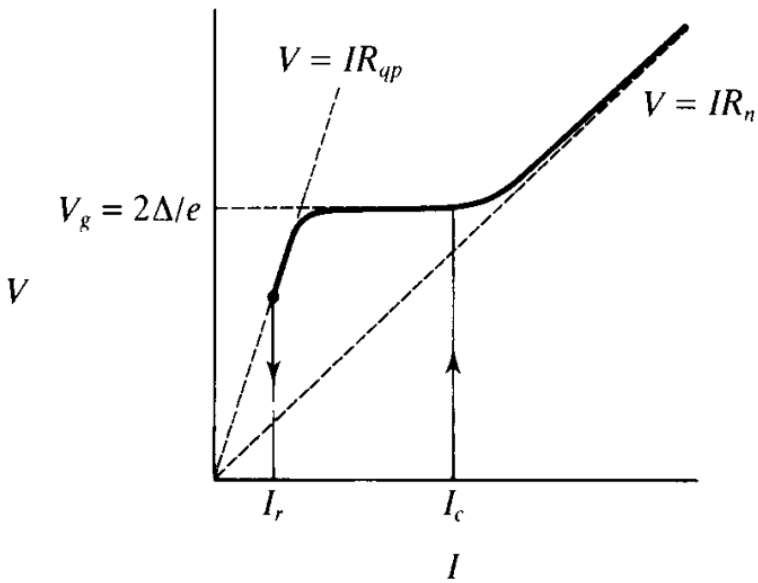

Figure 7: The graph shows the Hysterisis of the I-V curve for Under-damping junction

The figure shows that as the current increase there is no voltage until the current reaches the value of critical current, at this point the voltage V jumps discontinuously up to a finite voltage, this is called the running state, as the phase φ change by an increasing rate of $\frac{2eV}{\hbar}$, decreasing the current I from this point, the voltage don't follow it's initial path dropping to zero at I_c but take time until it reaches a trapping current I_r.

6 Dynamics of Three Josephson Junctions Connected in Series

The dynamics of junctions connected in parallel mimic coupled pendulums, despite the chaotic behavior of the system a synchronization for the chaotic motion can be observed as presented by James A .Blackburn et-al [3], the circuits presented in this paper uses an ac current source and coupled parallel-connected Josephson junctions, making a numerical simulation for the 2 junctions coupled parallel.
Advancing a small step further we take shunted three series-connected Josephson junctions as illustrated in Figure 8.

In this circuit the total current that passes through the junction is i_j:

$$i_j = i_0 - i_s \tag{40}$$

From the circuit diagram i_0 is the source current and i_s is the current through the shunted resistance, and the current through junctions is the same but as mentioned before the phase of the macroscopic wave function change after the tunneling, so the current i_j must satisfy Equation (30), also the voltage in each junction is related to the junctions phase similar to Equation (29), So for junction k as an example.

$$k = 1, 2, 3 \tag{41}$$

$$i_j = i_{c_k} sin\ \varphi_k + \frac{1}{R_k}\frac{\hbar}{2e}\frac{d\varphi_k}{dt} + C_k\frac{\hbar}{2e}\frac{d^2\varphi_k}{dt^2} \tag{42}$$

$$V_k = \frac{\hbar}{2e}\frac{d\varphi_k}{dt} \tag{43}$$

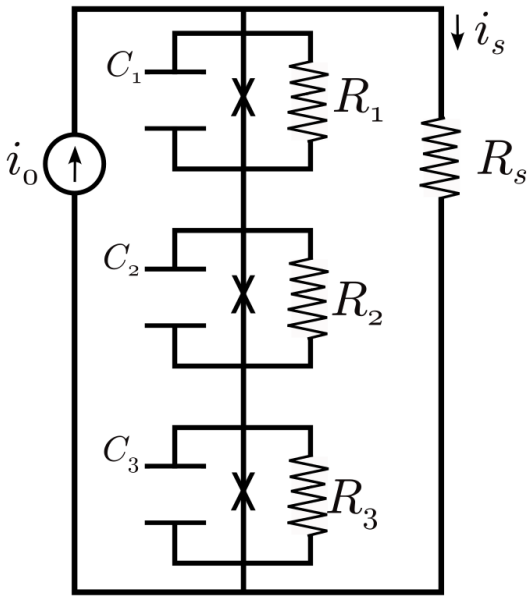

Figure 8: Three Josephson junctions connected in series and shunted by a resistor

V_k is the voltage across junction **k**, $\boldsymbol{i_{c_k}}$ is the critical current of junction **k**, φ_k is the phase of junction **k**, R_k and C_k are the resistance and capacitance of junction **k** respectively. The Voltage across the three junctions equals the voltage over the shunt resistance, equals the total voltageV_T.

$$V_T = \sum_{k=1}^{3} V_k = i_s R_s \tag{44}$$

So the current through the shunt resistance is

$$i_s = \frac{\hbar}{2eR_s}\frac{d}{dt}(\varphi_1 + \varphi_2 + \varphi_3) \tag{45}$$

and Equating Equation (40) & (42).

$$i_0 - i_s = i_{c_k} sin\ \varphi_k + \frac{\hbar}{2eR_k}\frac{d\varphi_k}{dt} + C_k\frac{\hbar}{2e}\ \frac{d^2\varphi_k}{dt^2} \tag{46}$$

Dividing the equation by i_{c_k}

$$i_0^* - \frac{i_s}{i_{c_k}} = sin\ \varphi_k + \frac{\hbar}{2ei_{c_k}R_k}\frac{d\varphi_k}{dt} + C_k\frac{\hbar}{2ei_{c_k}}\ \frac{d^2\varphi_k}{dt^2} \tag{47}$$

Normalizing the time scale $\tau = \omega_{J_k} t$ using the plasma frequency ω_{J_k} .

$$\omega_{J_k} = \sqrt{\frac{2ei_{c_k}}{\hbar C_k}} \tag{48}$$

so equation (45) & (47) becomes

$$i_s = \frac{\hbar\omega_{J_k}}{2eR_s}\frac{d}{d\tau}(\varphi_1 + \varphi_2 + \varphi_3) \tag{49}$$

$$i^*{}_0 - \frac{i_s}{i_{c_k}} = sin\varphi_k + \frac{1}{R_k}\sqrt{\frac{\hbar}{2ei_{c_k}C_k}}\frac{d\varphi_k}{d\tau} + \frac{d^2\varphi_k}{d\tau^2} \tag{50}$$

$i^*{}_0 = \frac{i_0}{i_{c_k}}$ is the normalized dimensionless current, and we define the RCSJ damping parameter β_J. Keeping in mind that applying this normalization to Equation (43) just means that the solution of v[t] is for time in units of plasma frequencyω_{J_k}

$$\beta_{J_k} = \frac{1}{R_k}\sqrt{\frac{\hbar}{2ei_{c_k}C_k}} \tag{51}$$

substituting in Equation (49)

$$i_s = i_{c_k}\frac{R_k}{R_s}\beta_{J_k}\ \frac{d}{d\tau}(\varphi_1 + \varphi_2 + \varphi_3) \tag{52}$$

Letting $\alpha_k = \frac{R_k}{R_s}\beta_{J_k}$ we reach the final form of our equations.

$$i^*{}_0 - \alpha_1\frac{d}{d\tau}(\varphi_1 + \varphi_2 + \varphi_3) = sin\varphi_1 + \beta_1\frac{d\varphi_1}{d\tau} + \frac{d^2\varphi_1}{d\tau^2} \tag{53}$$

$$i^*{}_0 - \alpha_2\frac{d}{d\tau}(\varphi_1 + \varphi_2 + \varphi_3) = sin\varphi_2 + \beta_2\frac{d\varphi_2}{d\tau} + \frac{d^2\varphi_2}{d\tau^2} \tag{54}$$

$$i^*{}_0 - \alpha_3\frac{d}{d\tau}(\varphi_1 + \varphi_2 + \varphi_3) = sin\varphi_3 + \beta_3\frac{d\varphi_3}{d\tau} + \frac{d^2\varphi_3}{d\tau^2} \tag{55}$$

$$\tag{56}$$

Assuming the junctions are identical and dropping the indexes of β & α
Having a look on the second term on the RHS, it's clear that as α increase the coupling between the three equations increase increasing the interaction between there phases. Setting some variables to solve the equation numerically.

```
Rs_1 = 0.0025;     (*the small value of shunted resistance*)
R = 0.001;     (*resistance of Josephson junctions as in RCSJ
   model*)
b = 0.25;    (*Damping parameter*)
α = (R/Rs_1)*b
α=0.1
```

Listing 1: Values used for solution

The very small value shows how we are using a shunted resistance.

7 Numerical Simulation

7.1 Mathematica Code

```
(*these sets are used in parametricNDSolve equation ir1 set shows the behaviour in the running state, ir2 shows the behaviour in the trapping state*)
ir1 = {1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9};
ir2 = {0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9};
(*defining the equations we solve*)
equ1 = ir- α (y1'[t] + y2'[t] + y3'[t]) == Sin[y1[t]] + b y1'[t] + y1''[t];
equ2 = ir- α (y1'[t] + y2'[t] + y3'[t]) == Sin[y2[t]] + b y2'[t] + y2''[t];
equ3 = ir- α (y1'[t] + y2'[t] + y3'[t]) == Sin[y3[t]] + b y3'[t] + y3''[t];
(*defining the initial conditions of function and first derivative*)
ini1 = y1[0] == 0;
ini2 = y2[0] == 0;
ini3 = y3[0] == 0;
dini1 = y1'[0] == 6*10^-15*(ℏ/2e)^-1;
dini2 = y2'[0] == 0;
dini3 = y3'[0] == 0;
(*Solving the equation for y1 (first junction phase) and ploting the solution *)
s01=ParametricNDSolveValue[{equ2, equ1, equ3, ini1, ini2, ini3, dini1, dini2, dini3}, y1, {t, 0, 500}, {ir}];
phase1 = Plot[Evaluate[Table[s01[ir][t], {ir, irvalues}]], {t, 0, 500}, PlotRange -> All,PlotLegends -> irvalues, AxesOrigin -> {0, 0}, AxesLabel -> {"t", "φ1"}]
(*Defining the Voltage function of the first junction to plot it later*)
V1[t] =(ℏ/2e) D[s01[ir][t], t];
(*lines from 15 to 19 are repeated for each junction to plot the phase and the voltage*)
(*Ploting the total voltage of the junctions over time*)
Plot[Evaluate[((V1[t] + V2[t] + V3[t]) /. ir -> # & /@
ir1], {t, 0, 500}, PlotRange -> All, PlotLegends -> irvalues,
AxesLabel -> {"t", "V_T[t]"}, AxesOrigin -> {0, 0}]
(*All the previous steps are repeated to for ir2 set to find the complete I-V curve*)
(*Then we select the values of voltage after some time at the steady state and list them to plot the I-V Curve*)
minvalues1 = Table[MinValue[{(V1[t]), 450 <= t <= 500}, t], {ir, ir1}];
maxvalues1 = Table[MaxValue[{(V1[t]), 450 <= t <= 500}, t], {ir, ir2}];
listy1 = Join [minvalues1, maxvalues1];
listx = Join [ir1, ir2];
P1v01 = ListPlot[Transpose[{listx, listy1}], Frame -> True , FrameLabel -> {"i*", "Volt"}, PlotStyle -> PointSize[Large],
```

```
   Joined ->True, AxesOrigin -> {0, 0},PlotLabel -> "I-V curve
   for j1 phase intial conditions 0, V1_0=6*10^-15*(ħ/2e)^-1 "]
(*this code plots the total voltage relation with current *)
minvalues = Table[MinValue[{(V1[t] + V2[t] + V3[t]), 450 <= t
   <= 500}, t], {ir, irvalues1}];
maxvalues = Table[MaxValue[{(V1[t] + V2[t] + V3[t]), 450 <= t
   <= 500}, t],{ir, irvalues}];
listy = Join [minvalues, maxvalues];
listx = Join [irvalues1, irvalues];
P02 = ListPlot[Transpose[{listx, listy}], Frame -> True ,
   FrameLabel ->{"i*", "Volt"}, PlotStyle -> Dashed, Joined ->
   True,AxesOrigin -> {0, 0}, PlotLabel -> "I-V curve for all
   Phase initial conditions is 0 "]
```

Listing 2: Mathematica notebook for Ploting the I-V curve of the three junctions

7.2 Behavior of the circuit: Graphical Simulation

Firstly, keep in mind that all the graph we present as an example to show the Circuits behavior and doesn't represent the resolution of values of current we used.
Also all time axes in these graphs are in units or plasma frequency.

7.2.1 The Phase Initial Conditions

In this section we change the initial conditions of the junctions phase and see the behavior of the junctions, Setting the initial conditions to zero.
$\varphi_1[t] = 0, \qquad \varphi_2[t] = 0, \qquad \varphi_3[t] = 0.$

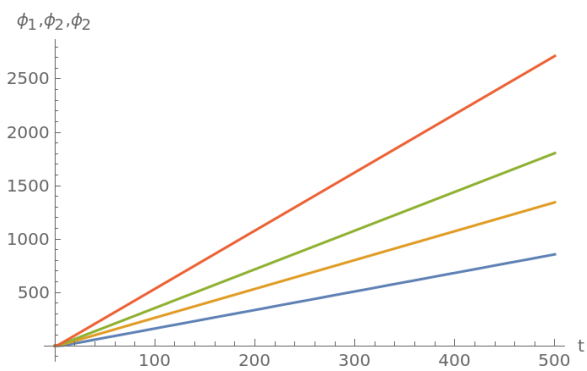

Figure 9: the three junctions have identical behavior

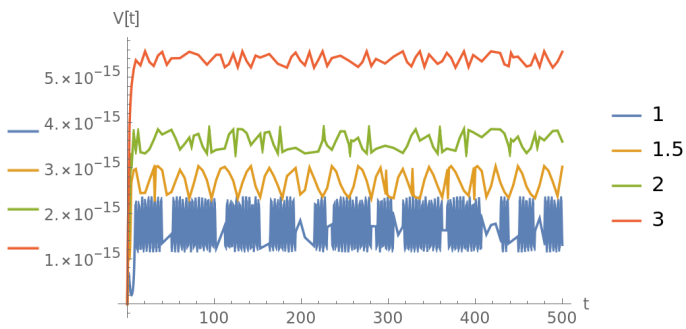

Figure 10: the variation of voltage is so small it can be neglected, also as the current increase the maximum voltage increase.

These where the solutions for currents larger than critical current, For current less than critical current we find the following.

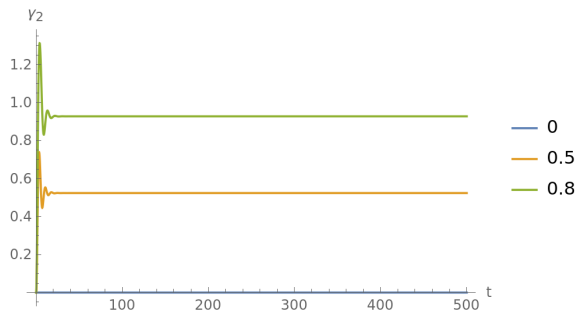

Figure 11: the phases less than critical current reaches a constant state unlike larger than critical current the phases increased indefinitely

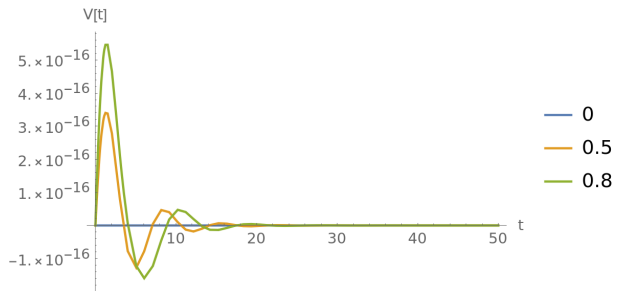

Figure 12: Voltage varies around zero until reaching a steady state of zero voltage

The I-V curve for this condition is simply the same as one Josephson junction.

These behaviors Shows that the tilted-washboard model is at hand, As the current increases the potential's slope increase pushing the particle in the positive direction of it's axes(φ here) to fall indefinitely increasing it's position. This allows us to neglect the behavior of the voltage and phase over time in the next examples and focus on the I-V Curve

For the following initial Conditions:

- $\varphi_1[t] = 0,\ \varphi_2[t] = \frac{\pi}{2},\ \varphi_3[t] = 0,$ RED Curve.
- $\varphi_1[t] = 0,\ \varphi_2[t] = \frac{\pi}{2},\ \varphi_3[t] = \frac{\pi}{2},$ GREEN Curve.

It's clear that the Junction shows a hysteresis property when changing the initial phase, But as the phase is not a quantitative physical quantity, We turn to studying the behavior for different phase first derivative initial conditions, Which is related to voltage.

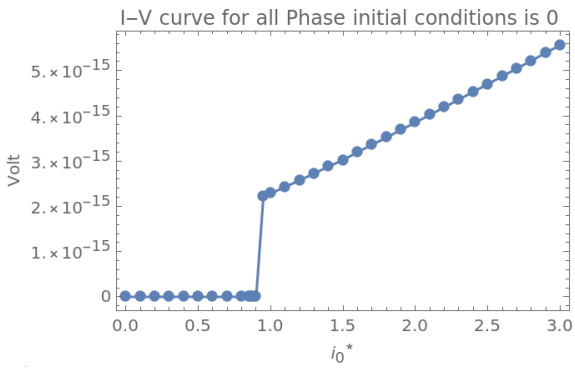

Figure 13: The points in this figure indicates the resolution of our calculation, and the points resulting from the simulation.

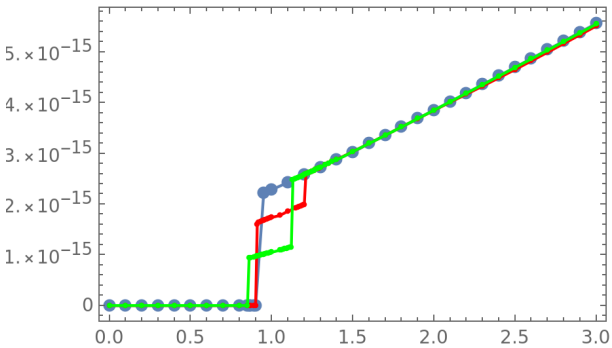

Figure 14:

7.2.2 The Voltage initial Conditions

In this section we change the initial conditions of the junctions Voltage and see the behavior of the junctions, as the initial condition to be all zero was presented before now we set the first junction to have an initial voltage, The Values of phase first derivative we choose hear are equal to the maximum value the junction reaches when it jumps discontinuously to the running state, the conditions are: $\varphi_1'[t] = (1\times10^{-15})\times(\frac{\hbar}{2e})^{-1}$, $\varphi_2'[t] = 0$, $\varphi_3'[t] = 0$, We multiply by $(\frac{\hbar}{2e})^{-1}$ as we are using the value of the phase derivative not the voltage. In the graphs it's clear that the junctions interact, such that the phases at specific current is posted while other wise for these currents the junction would be in the re-trapping state. The

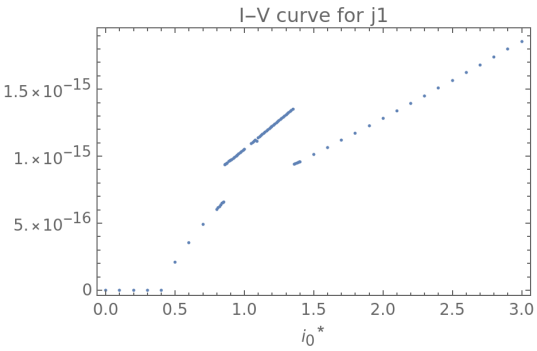

Figure 15:

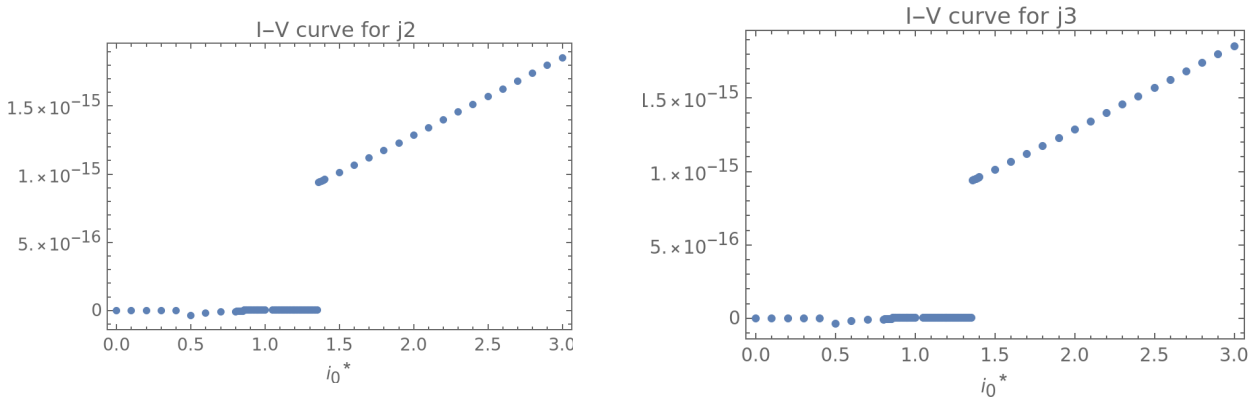

Figure 16: The other junctions retain there behavior

same behavior is observed when giving the same initial conditions for 2 junctions, So the hysteresis in the I-v curve is observed in the next graph shows that the voltage reach zero depending on its initial voltage for the junctions.

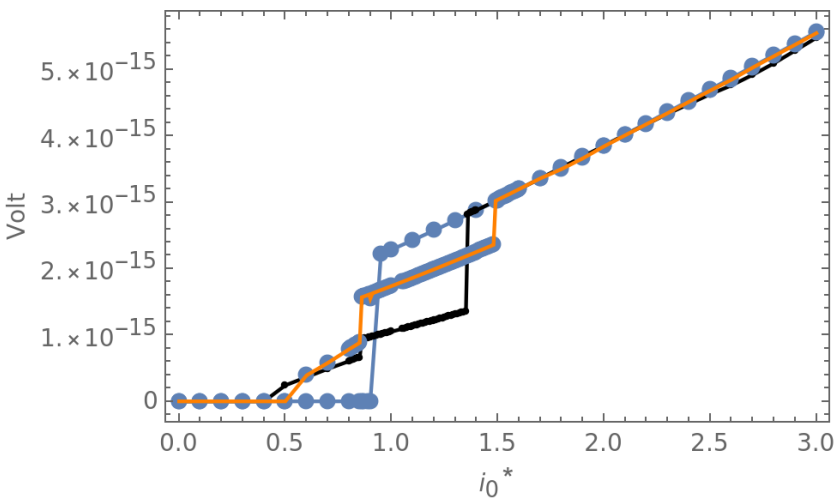

Figure 17:
a)The Black curve represents $\varphi_1'[t] = (10^{-15})(\frac{\hbar}{2e})^{-1}$,

b)The orange curve with blue points represents $\varphi_1'[t] = (10^{-15})(\frac{\hbar}{2e})^{-1}$, $\varphi_2'[t] = (10^{-15})(\frac{\hbar}{2e})^{-1}$

8 Conclusions & Remarks

A small remark that we observed in Figure 15 is that starting with the initial condition at a large value of voltage and current and decreasing the current, we find an abrupt increase in voltage like a step then the voltage the with decreasing current till reaching zero eventually, using the titled-washboard model and trying to plot this potential in 3 dimensions for only 2 phases, this potential shows an

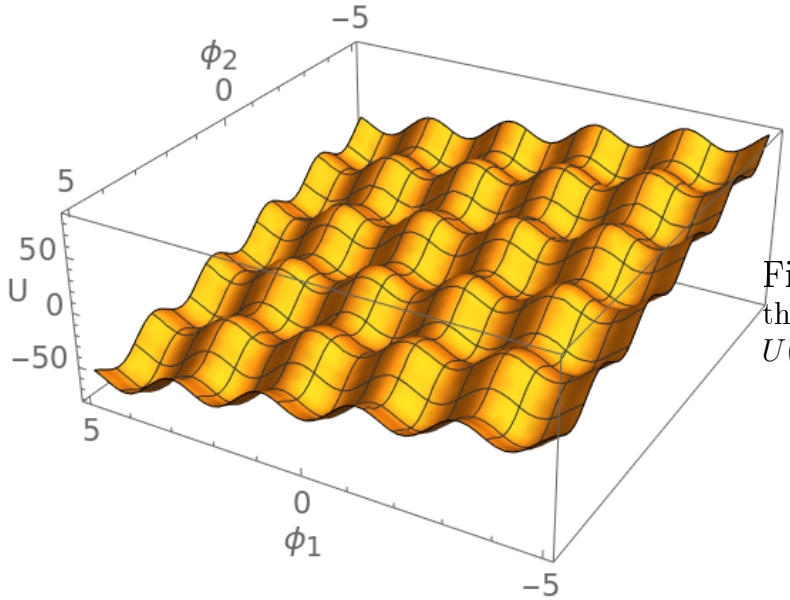

Figure 18:
the equation used is:
$U(\varphi_1, \varphi_2) = -cos\varphi_1 - \varphi_1 - cos\varphi_2 - \varphi_2$

inclined plane with small hills on it, such that the particle moving on it which is the total phase in this case can due to its initial position be slide in a certain direction instead of an other that caused that at currents less than the critical current the second and third junctions phases reached a constant phase while the first junction with a different initial phase (position on the plane) continued changing (running down the slope).
We show the change the can happen for such potential if we change the value of current. In Conclusion, using different resistances for the junction by changing the superconductor material or the type of links and bridges and the shunted resistance values can help us design various circuits that can be used in different purposes according to there behavior.

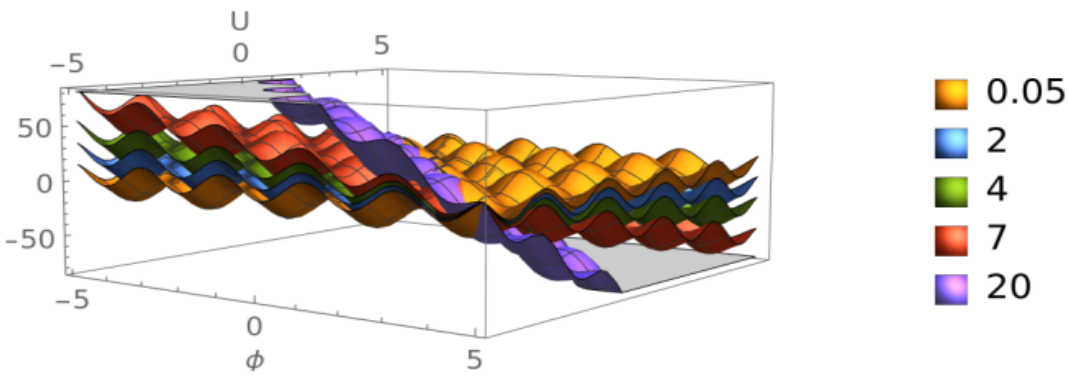

Figure 19: The Tilted-Washboard Potential for different values of current, largest current have the largest slop.

References

[1] Michael Tinkham, *Introduction to Superconductivity*, 2nd edition, McGraw-Hill, 1996.

[2] Christian Enss and Siegfried Hunklinger, *Low-Temperature Physics*, Springer, 2005.

[3] James A .Blackburn, Gregory L.Baker and H.J.T. Smith, *Intermittent Synchronization of Resistively Coupled Chaotic Josephson Junctions*, PHYSICAL REVIEW B, Vol 62, No.9 ,2000. DOI: 10.1103/PhysRevB.62.5931

[4] Gleb Krylov, Tahereh Jabbari, Eby G.Friedman, *Single Flux Quantum Integrated Circuit Design 2nd Ed*, Springer, 2024, https://doi.org/10.1007/978-3-031-47475-0

[5] Juan J.Mazo and Alexey V.Ustinov, *The Sine-Gordon Equation in Josephson-Junction Arrays*, Springer, 2014, DOI 10.1007/978-3-319-06722-3__7,